



Flying Liferaft

FLR05: Open Loop Dynamic Stability



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Group Design Project
H401 MEng Aeronautical Engineering

20/06/2022

Abstract

This report outlines the derivation of a state-space model used to predict the open loop response of the FREGATA flying Liferaft. There is a particular emphasis on how the methods used to calculate stability derivatives for conventional aircraft have been adapted to account for the unique features of FREGATA such as the H-tail empennage, pusher propeller and long, protruding engine pylons. The open loop response of each of the longitudinal and lateral modes is then analysed and results are compared to those obtained by the Flight Simulator team. Furthermore, investigations into varying aircraft centre of gravity position, fuel weight and free stream velocity are all presented. The results show that all of the aircraft modes are open loop stable although there are significant differences to the flight simulator data. Many of these differences can be attributed to the modelling assumptions as well as not being able to properly account for the effect of the pusher propeller. The report concludes with a series of recommendations made for the next design iteration.

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Nomenclature

x_{cg}	Centre of Gravity position along the x-axis
y_{cg}	Centre of Gravity position along the y-axis
z_{cg}	Centre of Gravity position along the z-axis
l_T	Moment arm of thrust relative to x-axis
U_e	Free-stream velocity
α	Angle of Attack
α_e	Equilibrium Angle of Attack
$\bar{c}$	Mean Aerodynamic Chord
L	Aircraft rolling moment
M	Aircraft pitching moment
N	Aircraft yawing moment
θ	Roll angle
q	Pitch angle
ψ	Yaw angle
p	Roll rate
q	Pitch rate
r	Yaw rate
K_n	Static Margin
C_L	Lift Coefficient
$(C_L)_e$	Equilibrium Lift Coefficient
M_∞	Free-stream mach number
c_s	Speed of Sound
C_{D0}	Zero Lift Drag Coefficient
l_H	x Distance between the horizontal tail centre of lift and the aircraft centre of gravity
h_H	z Distance between the horizontal tail centre of lift and the aircraft centre of gravity
β	side slip angle
β_s	slipstream side slip angle
C_{Yref}	Propeller Sideforce coefficient
S_P	Propeller disk area
C'_T	Propeller Thrust Coefficient
β_0	Pitch Angle
J	Propeller advance ratio
σ_e	Propeller solidity
x_s	distance from the propeller disk to the centre of the lift increment
z_{cp}	height of the centre of pressure of the vertical fin above the horizontal tail
z_F	height of the centre of pressure of the vertical fin above the aircraft x-axis
m_F	x-distance of the fin centre of pressure aft of the centre of gravity

1 Introduction

Open loop dynamic stability analysis is crucial during the conceptual aircraft design phase and can influence major design decisions such as centre of gravity location, wing and tail plane sizing, dihedral and sweep. Since this UAV's payload is merely a life-raft, passenger comfort is not a consideration but if any of the modes are found to be unstable, the aircraft could potentially plunge or spiral towards the sea and injure the very people that it is trying to save. Furthermore, if the UAV is prone to instability and crashes into the sea following significant perturbation to its states, the UAVs will need continual replacement and will dramatically increase overall project costs.

The first objective of this report is to detail the state-space model used to analyse the open-loop dynamic stability during cruise. The report then highlights how the 'unconventional' aspects of the design such as the H-tail empennage, propellers and protruding engine pylons have been taken into account when calculating stability derivatives. The final objective is to analyse the open loop dynamic response for each of the dynamic modes, paying particular attention to how the centre of gravity, cruise velocity and fuel weight affect the eigenvalues. Recommendations are highlighted in the conclusion section and are passed to the relevant sub teams.

2 State Space model

2.1 Aircraft states

The aircraft states include forces, moments and angular velocities concerned with the aircraft moving along or rotating about the three orthogonal axes (x, y and z). A generalized diagram showing aircraft body orthogonal axes are shown in Figure 1. All derivations in this report are done with respect to these and not earth fixed axes. All moments are taken about the aircraft centre of gravity of $x_{cg} = 616.3$ mm, $y_{cg} \approx 0$ mm and $z_{cg} = 3.2$ mm from the pusher centre line. The states are presented in Table 2 below with symmetric states (those affecting the aircraft in the x-z plane) being highlighted with an asterisk (*). These will form part of the aircraft's longitudinal equations of motion and will be analysed separately to the asymmetric states which form part of the aircraft's lateral equations of motion.

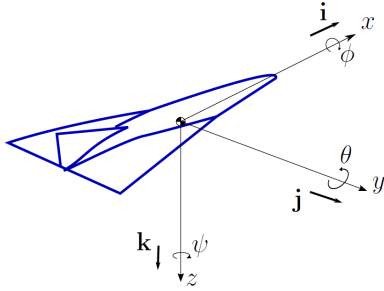


Figure 1: Aircraft orthogonal axes [1]

Table 2: Aircraft States

Axis	x	y	z
Force	X^*	Y	Z^*
Moment	L	M^*	N
Velocity	U^*	V	W^*
Angle	ϕ	θ^*	ψ
Angular Velocity	p	q^*	r

2.2 Aircraft Forces and Equations of Motion

The main longitudinal forces acting on the aircraft in the perturbed state is shown in Figure 2 below. Note that l_T is the moment arm of the thrust relative to the aircraft x-axis.

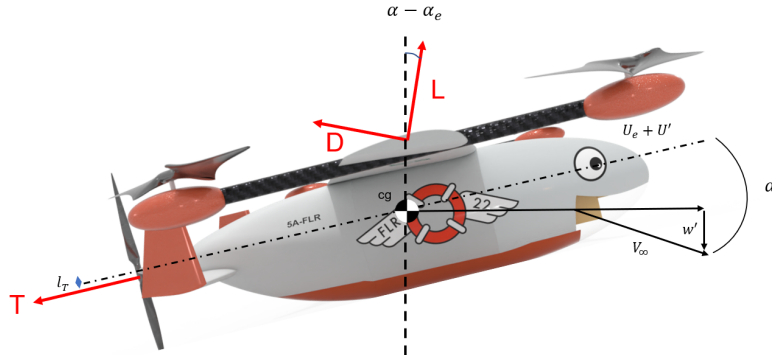


Figure 2: Aircraft forces in the Longitudinal Plane

By balancing forces in the X and Z direction and writing the lift and drag in terms of their respective non-dimensional coefficients,

$$X^a = T_x - \frac{1}{2}\rho U_e^2 S_{ref} [C_D \cos(\alpha - \alpha_e) - C_L \sin(\alpha - \alpha_e)] \quad (1)$$

$$Z^a = -T_z - \frac{1}{2}\rho U_e^2 S_{ref} [C_L \cos(\alpha - \alpha_e) + C_D \sin(\alpha - \alpha_e)] \quad (2)$$

where superscript a denotes 'aerodynamic' and subscript e denotes 'equilibrium conditions'. Following a static stability analysis presented in the report of Yi En Puah [2], the net moment acting on the aircraft is:

$$M^a = T l_T + \frac{1}{2}\rho U_e^2 S_{ref} \bar{c} [C_{M0} - K_n C_L] \quad (3)$$

where $\bar{c}$ is the mean aerodynamic chord, C_{M0} is the moment coefficient about the centre of gravity when $C_L = 0$ and K_n is the static margin. These equations were used in the derivations of the longitudinal stability derivatives discussed in Section 3.1.

The aircraft equations of motion from reference [1] were linearized about an equilibrium point where all the states are zero except for U which is equal to the free stream velocity of 58.3 m/s. The decoupling of the system into longitudinal and lateral equations of motion made use of the Curie Principle which assumes that asymmetric effects cannot result from a symmetric cause and vice versa [1]. The final linearized equations of motion in the longitudinal and lateral planes are shown in Equation 4 and Equation 5 below:

$$\begin{bmatrix} m & -X_{\dot{w}} & 0 & 0 \\ 0 & m & -Z_{\dot{w}} & 0 \\ 0 & -M_{\dot{w}} & I_{yy} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{U} \\ \dot{W} \\ \dot{q} \\ \theta \end{bmatrix} = \begin{bmatrix} X_U & X_W & X_q & -mg \\ Z_U & Z_W & Z_q + mU_e & 0 \\ M_U & M_W & M_q & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} U \\ W \\ q \\ \theta \end{bmatrix} \quad (4)$$

$$\begin{bmatrix} m & 0 & 0 & 0 & 0 \\ 0 & I_{xx} & -I_{xz} & 0 & 0 \\ 0 & -I_{xz} & I_{zz} & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{V} \\ \dot{p} \\ \dot{r} \\ \dot{\phi} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} Y_V & Y_P & Y_r - mU_e & mg & 0 \\ L_V & L_P & L_r & 0 & 0 \\ N_V & N_P & N_r & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} V \\ p \\ r \\ \phi \\ \psi \end{bmatrix} \quad (5)$$

Where the notation A_B denotes the derivative of state A with respect to state B. I_{xx} , I_{yy} , I_{xz} and I_{zz} are the aircraft moments of inertia with respect to the relevant aircraft body axis.

3 Stability Derivatives Methodology

In the interest of brevity, the full methods used to calculate the stability derivatives aren't detailed in this report. The main objective is to convey how standard methods of calculating these stability derivatives for conventional aircraft have been adapted for FREGATA. The main unconventional attributes of this aircraft include four vertical propellers, one pusher propeller, long protruding engine mounts and an H-tail empennage. When deriving stability derivatives and ultimately the open-loop model, the following assumptions were made:

1. Angles of attack are small and $\alpha = \alpha_e$
2. All mach number derivatives are assumed to be zero (since $M_\infty < 0.3$).
3. The lateral stability derivatives don't vary significantly with cg and velocity. These are mainly dependent on aircraft geometric and aerodynamic parameters.
4. All drag contributions of the tail plane and stationary propellers are assumed to be included in the aircraft total C_{D0} .
5. The thrust doesn't vary significantly with α and U_e , i.e $\frac{dT}{d\alpha} = \frac{dT}{dU_e} = 0$. This is valid due to small angles of attack and since assumed cruise velocity is more or less constant.
6. All perturbations to initiate the open loop modes are assumed to be reasonably small. The aircraft will not be designed to withstand large perturbations in open loop.

3.1 Longitudinal Stability Derivatives general methodology

The X and Z forces well as the moment M equations presented in Section 2.2 were used to derive the longitudinal stability derivatives as per reference [1]. These derivations are fairly standard hence are not explained in-depth in this report, however, their subsequent adaptation to account for engine pylons and propellers are discussed in Section 3.2.1 and Section 3.3.1. The exact formula's used in the calculations are shown in in Equation 25 to Equation 36 in Appendix B. Note that these equations M is the freestream mach number, c_s is the speed of sound, Subscript 'H' denotes horizontal tail plane. The moment arms $l'_H = x_H - x_{cg}$ and $h_H = z - z_{cg}$.

3.1.1 Lateral Stability Derivatives general methodology

The lateral stability derivatives for cruise were estimated using several empirical models from the ESDU [3]. Effects from the H-tail, fuselage and propeller pylons were considered as appropriate. Again, the full methodology is not included in this report - rather, an emphasis is made on the non-conventional components. A comprehensive list of the exact ESDU data sheets used for these derivations is shown in Table 8 in Appendix C. The main contributors to each derivative are highlighted in the Table 3 where w is the wing contribution, $T2F$ is due to the twin vertical fins, F is due to the fuselage body, ep is due to the engine pylons, ϵ is due to twist and P is for the pusher propeller. Note that all derivatives calculated using empirical models from ESDU had to be dimensionalized before incorporating them into the state space model.

Table 3: Lateral Stability Derivative Contributors

Side Force Derivatives		Rolling moment derivatives		Yawing moment derivatives	
Derivative	Contributors	Derivative	Contributors	Derivative	Contributors
Y_v	w, T2F, py	L_v	w, F, T2F, P, py	N_v	w, T2F py
Y_p	w, T2F	L_p	w	N_p	w, T2F
Y_r	T2F, F	L_r	w, ϵ , T2F	N_r	w, T2F, F

3.2 Effect of engine pylons

3.2.1 Effect of engine pylons on Longitudinal Stability derivatives

A somewhat distinct feature of the aircraft are the four engine pylons that protrude from the wing. These are a significant drag contributor and can cause interference of flow over the wing and the horizontal tail. In steady level cruise flight when the free stream velocity vector is nearly aligned with the aircraft body x-axis ($\alpha_e \approx 0$), the drag contribution in the x-z plane is accounted for in the total drag coefficient C_{D0} .

When the aircraft is pitching, there will be an induced vertical velocity W_i that varies along the length of the aircraft [1]:

$$W_i = (x - x_{cg})q \quad (6)$$

There will thus be a drag force acting in the vertical direction contributing to the overall Z force that depends on the pitch rate. To estimate this drag, the engine mounts were modelled as circular cylinders of diameter 34 mm. An estimated pitch rate of 45 degrees per second was used to obtain an induced vertical velocity of $W_i = 0.205$ m/s. Obviously, this pitch rate depends on the conditions and can vary significantly depending on gusts and wind shear, but for now this average value was taken from the Flight Dynamics and Control Laboratory in 2nd Year [4].

The Reynolds number of the flow is approximately equal to 460. According to experimental data from [5] shown in Figure 21 in Appendix A, the drag coefficient at $Re = 460$ is approximately 0.9. This was then accounted for when calculating the Z_W , Z_q and M_q derivatives which are all dependent on the total drag coefficient. It is worth noting that presence of these engine pylons lead to a 10.7 % increase in damping ratio of the SPPO mode and is thus deemed beneficial in this regard.

3.2.2 Effect of engine pylons on Lateral Stability derivatives

When the aircraft is in yaw with side slip angle β , the velocity component acting in the Y direction is $U_e \sin(\beta) \approx U_e \beta$ for small side slip angles. In order to estimate the effect of the pylons on the stability derivatives, an existing empirical method from [6] was adapted. This method is for engine nacelles (not pylons) but was implemented by treating each engine pylon as an extended engine nacelle. In theory, it would have been possible to treat the engine pylons as circular cylinders and investigate how the increase in drag affects the stability derivatives (as done in Section 3.2.1), however, the use of the empirical method was preferred for the following reasons:

1. Since the side slip angle affects the velocity which in turn affects the drag force experienced by the circular cylinder, the side slip angle would have had to be estimated but this varies depending on the conditions.
2. The empirical method accounts for interference effects.
3. Most lateral stability derivatives that will be calculated use empirical models so this is deemed somewhat consistent.

A potential drawback of this, however, is that an engine nacelle is typically a streamlined shape and the engine pylon (idealised as a circular cylinder) is not. The equation used to estimate the engine pylon contributions to the Y_v is shown in Equation 7 below:

$$(Y_v)_{py} = -\left(\frac{2}{3}\right)K\pi \frac{(d^2)_{py}}{S_{ref}} \quad (7)$$

where $d_{py} = 0.034m$ is the engine pylon diameter and K is the fineness ratio correction factor based on the length to diameter ratio. The result was then multiplied by four to account for the number of engine pylons. The contribution to the N_v derivative was then estimated using the following empirical relation from the same paper [6]:

$$(N_v)_{py} = (Y_v)_{py} \frac{m_0 - 0.46l_{py}}{b} \quad (8)$$

where m_0 is the distance from the end of the engine pylon to the aircraft cg and l_{py} is the length of the engine pylon. This formula was applied to each of the four engine pylons.

The engine pylon contribution to the L_v derivative was found based on the vertical distance of the engine pylon above the aircraft cg z_{py} :

$$(L_v)_{py} = \frac{-z_{py}}{b} (Y_v)_{py} + 0.86(L_v)_{pyT} \quad (9)$$

where $(L_v)_{pyT}$ takes into account the anti symmetric incidence loading due to engine mounts in side slip [6]. The engine pylons likely also affect the aircraft in roll and could have an impact on the L_p derivative. After consulting the literature [7], it was decided that the engine mount contribution would be neglected for this derivative. This is primarily because the planform area of the wings is much larger than the area of the engine pylons subject to the freestream velocity during roll, hence the wings provide far greater damping in roll than the engine pylons do.

3.3 Effect of Propellers

During cruise, the four vertical propellers do not spin and simply act as parasitic drag. These are taken into account in the high aircraft C_{D0} . The pusher propeller, however, affects both the longitudinal and lateral stability derivatives.

3.3.1 Effect of Propellers on Longitudinal Stability Derivatives

Propellers primarily affect stability derivatives due to the effect of the wing and empennage in the wake of the propeller, known as the slipstream effect [8]. For this particular aircraft configuration, the pusher propeller is aft of the wings and empennage and is hence out of the wake of the propellers. Nevertheless, the pusher propeller is still expected to affect the airflow over the H-tail and wing surfaces.

There is a plethora of literature available for tractor propellers that sit forward of the wing, however, there is limited data (be that experimental or empirical) to quantitatively assess the impact of a single, rear mounted pusher propeller on the stability derivatives. Presence of a pusher propeller causes increased suction over the wings leading to the separation point moving downstream and an increase in the wing lift coefficient ΔC_L and pressure drag [9]. Experimental data from [9] for C_L vs α and C_D vs C_L for a number of different pusher propeller configurations is shown in Figure 3 and Figure 4 respectively.

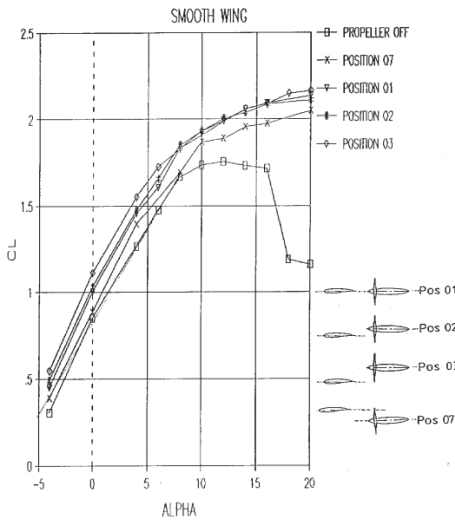


Figure 3: Experimental data for effect of propeller on C_L vs α

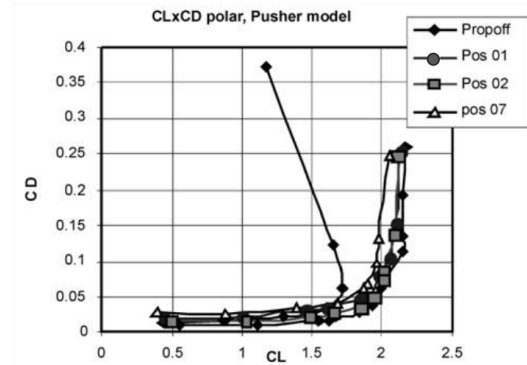


Figure 4: Experimental data for effect of propeller on Lift-Drag polar

In Figure 3, the configuration that most closely resembles our aircraft is 'Pos 07' (indicated with crosses 'x' in Figure 3) as this has a wing mounted higher than the propeller behind it. Based on this experimental data, ΔC_L is estimated to be +10.0 % and ΔC_D +250.0 %. There is a large amount of uncertainty in this estimate as the exact aerofoil of the wing studied and the distance between the wing and the propeller is not made clear in the paper. The C_L and C_D were then adjusted accordingly for stability derivative calculations.

3.3.2 Effect of Propellers on Lateral Stability Derivatives

When the aircraft is in sideslip, the propeller will generate a side force in its plane acting down-wind [8]. The effect of this sideforce can be quantified by calculating $\frac{dC_{Yref}}{d\beta}$ as shown in Equation 10:

$$\frac{dC_{Yref}}{d\beta} = -G \left(\frac{8}{\pi J^2} \right) \left(\frac{S_P}{S} \right) \quad (10)$$

where C_{Yref} is the propeller sideforce coefficient and β is the side slip angle. S_P is the propeller disk area and G is the side force derivative function which is related to the propeller thrust coefficient C'_T , pitch angle β_0 , advance ratio J and solidity σ_e by Equation 11 below:

$$G = \left(\frac{4.25\sigma_e}{1 + 2\sigma_e} \right) \sin(\beta_0 + 3) \left[\frac{\pi J^2}{8} + \frac{3 * (\frac{\pi J^2}{8})^{0.5} C'_T}{(8(\frac{\pi J^2}{8}) + (2/3)C'_T)^{0.5}} \right] \quad (11)$$

Doing this calculation resulted in a $\frac{dC_{Yref}}{d\beta}$ of 3.4×10^{-3} . By dimensionalizing and using chain rule, an approximate side force of 0.58 N was calculated which was considered negligible compared to other forces acting in the Y-plane such as the vertical fin force.

Furthermore, when the aircraft is sideslipping, the centre of incremental lift shifts. Assuming that a positive sideslip angle β is when the aircraft is crabbing starboard (right), the centre of incremental lift is displaced leftwards on both of the aircraft wings, thus causing a lift imbalance and a rolling moment.

The following method has been adapted from [8] for this aircraft configuration. The increment in rolling moment coefficient (ΔC_l) on the starboard and port sides of the aircraft can be calculated using Equation 12 and Equation 13 respectively:

$$(\Delta C_l)_{s_{starboard}} = -[(\Delta C_L)_s / 2] \frac{(y_p - x_s \beta_s)}{b} \quad (12)$$

$$(\Delta C_l)_{s_{port}} = [(\Delta C_L)_s / 2] \frac{(y_p + x_s \beta_s)}{b} \quad (13)$$

Where x_s is the distance from the propeller disk to the centre of the lift increment and β_s is the slipstream side slip angle. Adding $(\Delta C_l)_s$ on the starboard and port side and using chain rule gives the following result:

$$(L_v)_P = (\Delta C_l)_s / \beta = \Delta C_L)_s (x_s / b) (d\beta_s / d\beta) \quad (14)$$

3.3.3 Effect of H-tail Empennage

The aircraft design features an H-tail empennage with vertical fin height $h = 0.24$ m and horizontal span $b_T = 0.74$ m. The presence of an H-tail prevents the airflow spilling over from the pressure side to the suction side of the aerofoil for both the horizontal and vertical stabilizer [10]. This is known as end-plate effect and causes an increase in the effective aspect ratio of the fins. For an H-tail configuration with the horizontal tail directly below the vertical ones, the fin aspect ratio can be estimated to increase by factor 1.5 [11]. The centre of pressure of each vertical fin above the horizontal tail, z_{cp} , was assumed to be $0.424 h_{fin}$ from reference [11]. This corresponded to a height of $z_F = 0.632$ m above the aircraft body x-axis as shown in Figure 5. The x-distance of the centre of pressure of the fin behind the centre of gravity is denoted as m_F and has a value of 0.676 m.

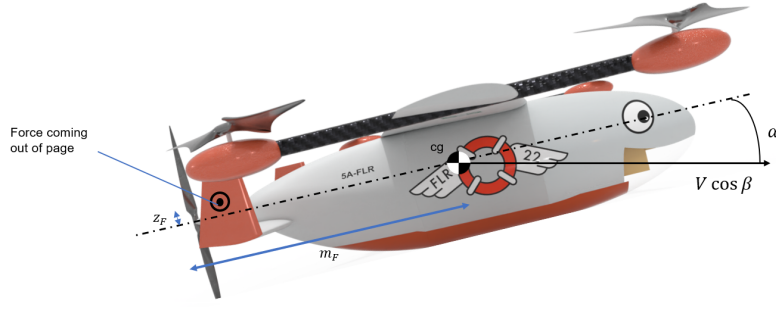


Figure 5: Geometric Dimensions relevant for stability derivative calculations

The $(Y_v)_{T2F}$ derivative was calculated using Equation 15 from [11] where S_F is the fin area and a_1 is the side force curve slope of the vertical fin.

$$(Y_v)_{T2F} = \frac{-2a_1 S_F}{S_W} \quad (15)$$

Calculation of the $(Y_r)_{T2F}$ relied on the key assumption that the yawing motion causes a sideslip velocity acting on the fin equal in magnitude to the yaw rate times the moment arm of the side force on the fin [12]. Using Figure 5 and reference [12],

$$(Y_r)_{T2F} = -(Y_v)_{T2F} \frac{(m_F \cos(\alpha) + z_F \sin(\alpha))}{b} \quad (16)$$

Then, by a similar moment balance about the appropriate axes, the fin contributions for N_r and L_r are given in Equation 17 and Equation 18 below [12]:

$$(N_r)_{T2F} = -(Y_r)_{T2F} \frac{(m_F \cos(\alpha) + z_F \sin(\alpha))}{b} \quad (17)$$

$$(L_r)_{T2F} = \frac{(Y_r)_F (z_F \cos(\alpha) - m_F \sin(\alpha))}{b} \quad (18)$$

The Sideforce due to sideslip acting at the centre of pressure on the vertical stabilizer generates a yawing moment about the aircraft z-axis. Calculation of the $(N_v)_{T2F}$ derivative involved a moment balance using the sideforce derivative $(Y_v)_{T2F}$ found previously [12]:

$$(N_v)_{T2F} = -(Y_v)_{T2F} \frac{(m_F \cos(\alpha) + z_F \sin(\alpha))}{b} \quad (19)$$

Similarly, the side force generates a rolling moment about the body x-axis. The corresponding contribution to the L_v derivative was found using Equation 20 from [12]

$$(L_v)_{T2F} = \frac{(-Y_v)_{T2F} (z_F \cos(\alpha) - m_F \sin(\alpha))}{b} \quad (20)$$

The vertical fins in side slip encounter anti symmetric loading over the tail plane. This is accounted for by adding an additional term, ΔL_v , to $(L_v)_{T2F}$. This term was estimated to be -0.035 using a semi-empirical method from [11] based on vortex-lattice calculations. The graphs used to estimate this value are shown in Figure 22 and Figure 23 in Appendix D.

4 Results and Discussion

The longitudinal and lateral equations of motion, Equation 4 and Equation 5 respectively, were recast into the form $\dot{x} = Jx$ where J is the Jacobian. The eigenvalues of this matrix were analysed in order to characterise the open-loop response of the aircraft.

4.1 Longitudinal Dynamic Stability

4.1.1 Longitudinal Stability Derivatives

The longitudinal stability derivatives calculated in dimensional form are shown in Table 4:

Table 4: Longitudinal Stability Derivatives

X-Force Derivatives		Z-Force derivatives		M Moment derivatives	
Derivative	Value	Derivative	Value	Derivative	Value
X_u	-8.321 Ns/m	Z_u	-18.991 Ns/m	M_u	0.000 Ns
X_w	9.391 Ns/m	Z_w	-6.217 Ns/m	M_w	-0.096 Ns
X_q	0.00 Ns	Z_q	-0.432 Ns	M_q	-0.292 Nms
$X_{\dot{w}}$	0.00 N s ² /m	$Z_{\dot{w}}$	-0.018 N s ² /m	$M_{\dot{w}}$	-0.012 N s ²

4.1.2 Open Loop Dynamic Response in cruise

The longitudinal open loop response was simulated in Simulink using the model presented by Bojun Yang [13] with all control gains set to zero. The SPPO mode was triggered by applying a doublet input (setting the elevator to +1 then -2 then +1 again). The idea was to perturb the pitch rate and not alter the pitch angle or velocity. The phugoid mode was triggered by applying a smaller positive elevator deflection (0.75) then the same corresponding negative elevator deflection instantaneously. The variation of the longitudinal states following these perturbations are shown in Figure 6 and Figure 7 for the SPPO and Phugoid modes respectively.

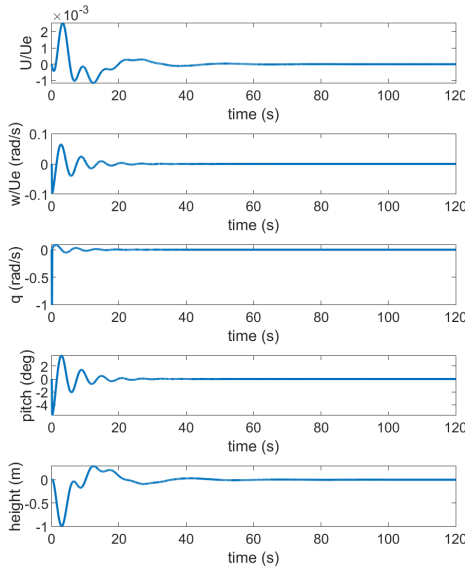


Figure 6: Open Loop: SPPO mode

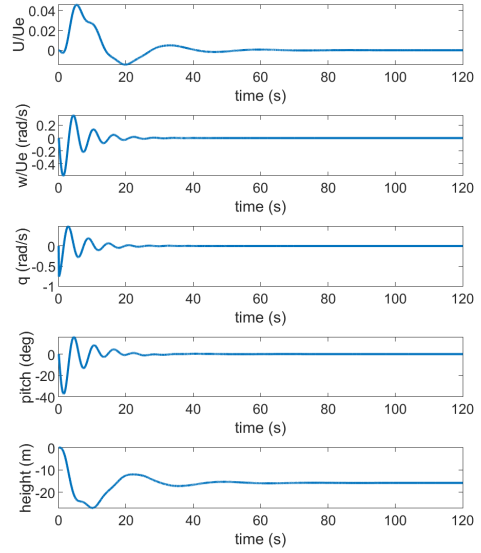


Figure 7: Open Loop: Phugoid mode

The eigenvalues of each dynamic mode as well as the damped natural frequency ω_d , undamped natural frequency ω_n , period T , damping ratio ζ and time to half amplitude $t_{1/2}$ is shown in Table 5. These are directly compared to data obtained from the flight simulation team in the same table.

Table 5: Longitudinal Dynamic mode characteristics

Mode	Test	Eigenvalues	ω_d (s ⁻¹)	ω_n (s ⁻¹)	T (s)	ζ	$t_{1/2}$ (s)
SPPO	Simulink	$-0.161 \pm 1.439i$	1.44	1.45	4.4	0.1113	4.3
	Flight Test	$-2.160 \pm 6.670i$	6.67	7.01	0.9	0.3074	0.3
Phugoid	Simulink	$-0.077 \pm 0.237i$	0.24	0.25	26.5	0.3084	9.0
	Flight Test	$-0.010 \pm 0.250i$	0.25	0.25	24.8	0.0424	49.1

On comparing Figure 6 and Figure 7 it can be seen that the SPPO mode does have a short period and results in a relatively smaller height perturbation to the aircraft, as expected. One particularly striking feature is that the SPPO and Phugoid modes seem to dampen out relatively quickly. Theoretically, the phugoid mode should be very lightly damped and so presence of this feature highlights the fact that there may be an issue with the longitudinal model. Table 5 shows that the predicted damping ratio of the Phugoid mode is in fact larger than the SPPO mode which should not be the case. The percentage difference in damping ratio between the predicted and flight test response is 63.0 % and 627.4 % respectively.

One potential reason for the large discrepancy in damping ratio for the SPPO mode could be the use of an inaccurate static margin and neutral point. The Flight Simulator team quoted the neutral point to be 0.623 m [ref] instead of 0.661 m predicted by the [2] doing the static stability analysis. If the eigenvalues are

recalculated using the flight simulator's neutral point, the damping ratio for the SPPO and Phugoid modes become 0.3025 and 0.2733 respectively. This results in a significant improvement in the predicted SPPO mode damping ratio with a percentage difference of only 1.6 %.

An analysis into a potential reason for the discrepancy seen with the Phugoid mode was conducted. Assuming that this mode is dominated by the X-force equation and since $Z_W \ll m$ and $X_u \ll Z_W$ for this UAV, the calculation of the eigenvalue corresponding to the Phugoid mode can be approximated as follows from [1]:

$$\lambda = -\frac{\gamma}{2} \pm \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{2g^2}{U_e^2}} \quad (21)$$

where

$$\gamma = \frac{\rho U_e S_{ref}}{2m} \left[\left(\frac{\partial C_L}{\partial \alpha} \right)_e + (C_D)_e \right] \quad (22)$$

Using this approximation yielded an eigenvalue of $-0.0606 \pm 0.2301i$ and a damping ratio of 0.2546 which differs from the state space calculated eigenvalue by 25%. Thus, analysing the eigenvalues with this approximation was deemed appropriate. By varying parameters such as U_e , $(C_D)_e$ and ρ , it was found that the term that most significantly contributes to the size of the eigenvalue, and in turn the damping ratio, is U_e . Data from the flight simulation indeed shows that flight tests were conducted at ≈ 54 m/s which is lower than the design cruise speed. Furthermore, the presence of a non-zero angle of attack could have lessened the effective free stream velocity even further. Another potential reason for the differences is the density ρ term. Tests were conducted at an altitude of nearly 600 m which is far greater than the design cruise altitude of 341 m. Finally there is a fair amount of uncertainty in the theoretical drag predictions which affects the C_{De} term.

4.1.3 Varying centre of gravity

The longitudinal stability derivatives were evaluated for a range of cg. This was done before the weight and balance team had finalized the layout of aircraft systems and components and served the purpose of giving the weight and balance team a dynamic stability constraint for cg. This could also be useful information to know if the aircraft is adapted for other military or civilian missions and requires a different kind of payload. The resulting eigenvalues for the SPPO and Phugoid modes with varying cg are shown in the root locus plots in Figure 8 and Figure 9 respectively. Note that in the figure 'E1' denotes eigenvalue 1 and 'E2' denotes eigenvalue 2.

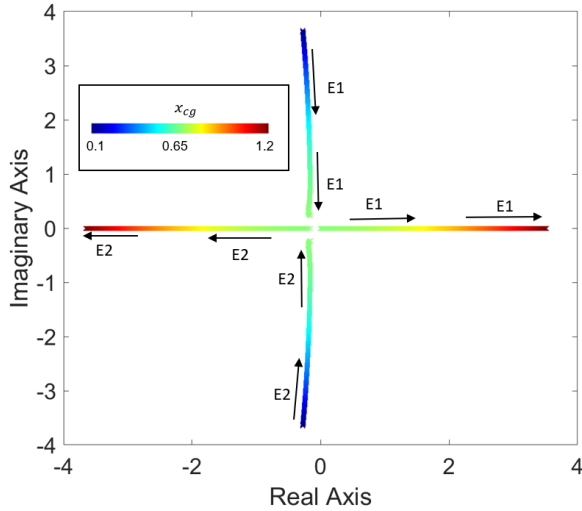


Figure 8: Root Locus Diagram varying x_{cg} (SPPO)

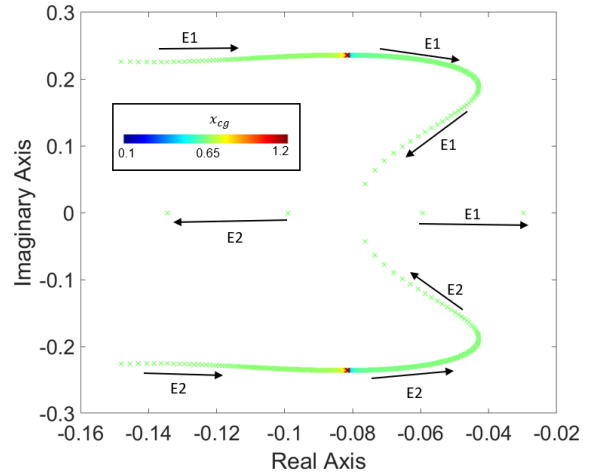


Figure 9: Root Locus Diagram varying x_{cg} (Phugoid)

Based on this, an upper bound of $x_{cg} = 0.6538$ m was passed to the weight and balance team. Beyond this centre of gravity position, the SPPO mode has an eigenvalue with a positive real part and is likely unstable. The same procedure was repeated for the Phugoid mode eigenvalues and the results are shown in Figure 9. It is evident from Figure 9 that is no value of x_{cg} in the range that renders the Phugoid mode unstable. In fact, it is noted that x_{cg} has very little effect on this mode - most of the eigenvalues are located at approximately $-0.08 \pm 0.24i$ and the variation in eigenvalues shown occurs only at x_{cg} of around 0.620 - 0.660 m. It is quite difficult to illustrate the eigenvalue behaviour on a 2D plot and so a 3D plot is shown in Figure 24 Appendix E where the x_{cg} value on the z-axis. An optimal x_{cg} of 0.660 gives the highest damping ratio but it was decided that this wasn't significant enough to drive the cg position. The weight and balance team later calculated the centre of gravity to be at 0.614 m [14].

4.1.4 Varying Fuel Weight

As fuel is used up, the centre of gravity shifts forward and the total aircraft mass decreases. The centre of gravity with zero fuel weight was found to be 0.593 according to the weight and balance team [14]. The effect of the eigenvalues on this change in centre of gravity due to change in fuel weight ($x_{cg,F}$) for the SPPO and Phugoid modes is shown in Figure 10 and Figure 11 respectively.

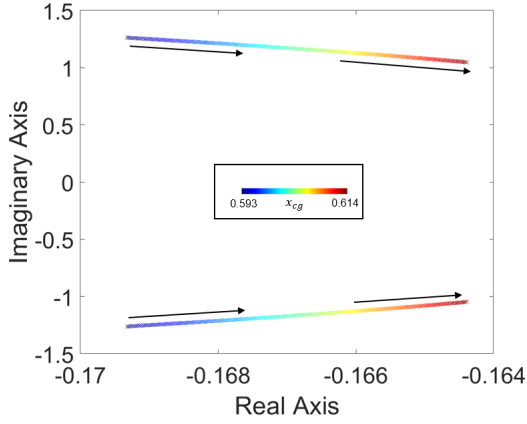


Figure 10: Root Locus for varying $x_{cg,F}$ (SPPO)

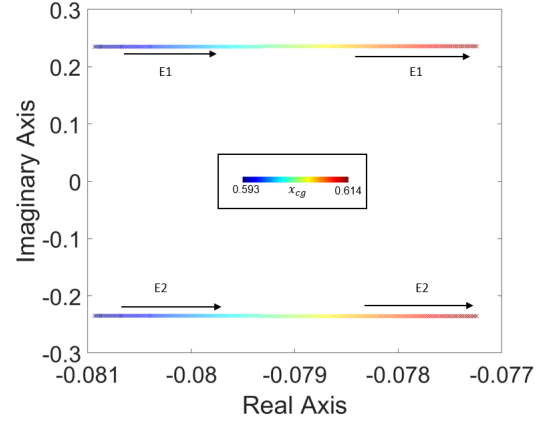


Figure 11: Root Locus for varying $x_{cg,F}$ (Phugoid)

It is evident that the change in fuel weight doesn't affect the stability characteristics of either mode.

4.1.5 Effect of Payload Drop

Although the payload is placed as close to the aircraft centre of gravity as possible, there will still be an instantaneous change in mass when the payload is dropped. The effect on the eigenvalues of the SPPO and Phugoid modes is shown in Figure 12 and Figure 13.

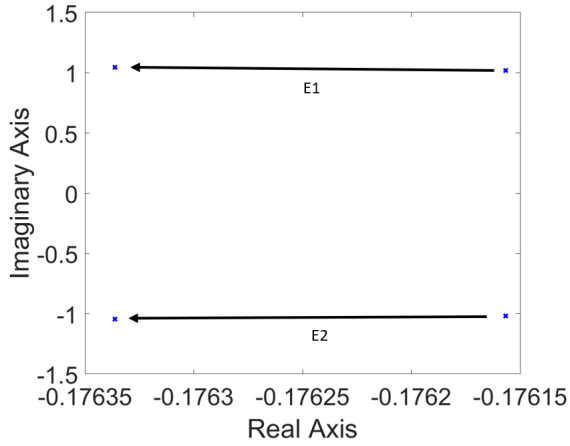


Figure 12: Payload drop effect (SPPO)

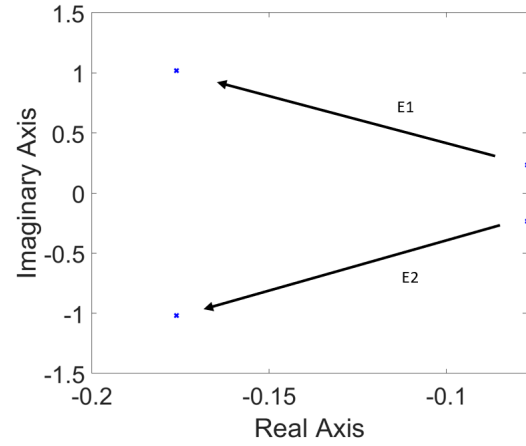


Figure 13: Payload drop effect (Phugoid)

The instantaneous change in mass has little effect on the eigenvalues of the SPPO mode. It does, however, decrease the damping ratio of the phugoid mode by 45.0 %. This finding could be particularly worrying since the phugoid mode was already very lightly damped. It is recommended that the payload drop is tested in the flight simulator in the near future to fully understand its effect on the open loop response.

4.1.6 Varying cruise velocity U_e

The effect of changing free stream velocity from the aircraft stall speed (30 m/s) to just over the aircraft cruise speed (65 m/s) was investigated and the results are presented in Figure 14 and Figure 15 below:

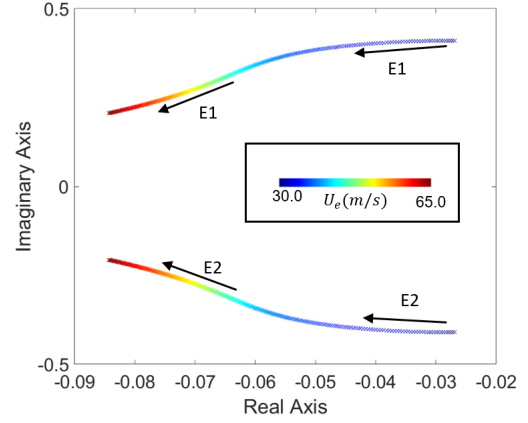
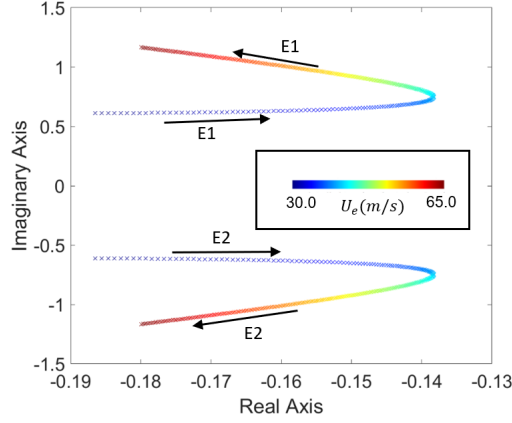


Figure 14: Root Locus Diagram varying U_e (SPPO) Figure 15: Root Locus Diagram varying U_e (Phugoid)

Firstly, it is noted that none of the velocities in the given range cause an instability of either of the dynamic modes. It is also noted that the damping ratio of the SPPO mode can be elevated by decreasing the cruise velocity below the 58.3 m/s. The general trend regarding the phugoid mode is that the damping ratio increases as U_e increases. Based on data collected by the flight simulator team, it is more important to prioritize increasing damping ratio of the phugoid mode over that of the SPPO mode which would mean increasing the cruise velocity. According to the SEP plot presented in [15], increasing the cruise velocity is currently not possible, but if propulsive or drag characteristics of the aircraft change in future iterations, it is worth considering. The results shown in Figure 15 are in agreement with Lanchester's Phugoid Approximation which suggests that the imaginary part of the Phugoid eigenvalue decreases as U_e increases [1].

4.2 Lateral Dynamic Stability

4.2.1 Lateral Stability Derivatives

The lateral stability derivatives calculated are presented in Table 6.

Table 6: Lateral Stability Derivatives

Side Force Derivatives		Rolling moment derivatives		Yawing moment derivatives	
Derivative	Value	Derivative	Value	Derivative	Value
Y_v	-11.14 Ns/m	L_v	-4.92 Ns	N_v	1.92 Ns
Y_p	0.59 Ns	L_p	-23.60 Nms	N_p	-0.32 Nms
Y_r	4.57 Ns	L_r	5.74 Nms	N_r	-4.95 Nms

4.2.2 Open Loop Lateral Response

Using the Lateral Simulink model from [13] with all control gains set to zero, the Lateral Open Loop dynamic response for each of the lateral dynamic mode was obtained. The results are shown in Figure 16, Figure 17 and Figure 18 (on the next page) for the Dutch Roll, Roll Subsidence and Spiral modes respectively. Furthermore, a table showing the characteristics of each dynamic mode is shown in Table 7. Due to difficulties associated with open loop control, the flight simulator team only tested Dutch Roll and hence a comparison is only made for that particular mode.

Table 7 shows that the Dutch Roll, Roll Subsidence and Spiral modes all have eigenvalues in the left half plane and are thus stable. Note that the eigenvalue of 0 corresponds to the Yaw independence mode. The physical implication of this is that the aircraft is neutrally stable with respect to heading when the aircraft's yaw angle is perturbed [1]. The predicted dutch roll mode period is sufficiently close to that obtained in the flight simulator tests but the damping ratio is significantly over-estimated, although not to the extent seen in the lateral modes.

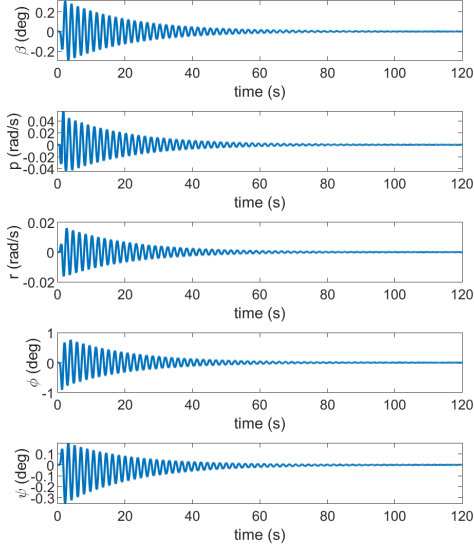


Figure 16: Open Loop: Dutch Roll mode

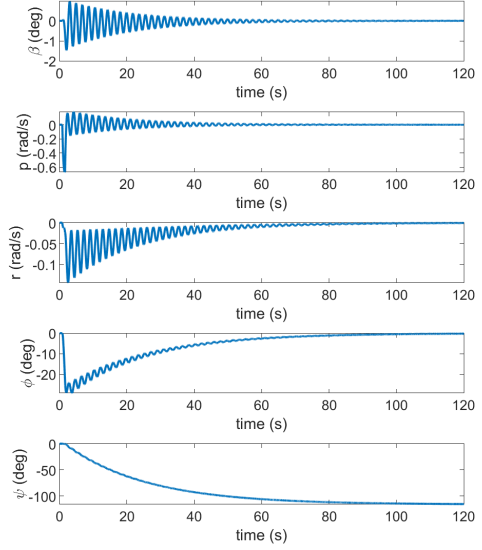


Figure 17: Open Loop: Roll Subsidence mode

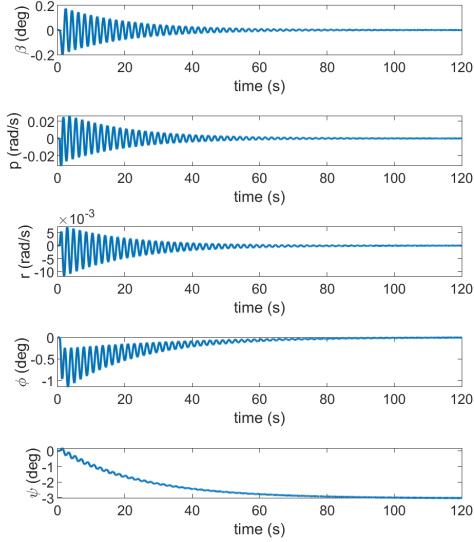


Figure 18: Open Loop: Spiral mode

Table 7: Lateral Dynamic mode characteristics

Mode	Eigenvalues	$T(s)$	ζ	$t_{1/2} (s)$
Yaw Independence	$0.000 + 0.000i$	0.0	0.0000	0.0
Dutch Roll	$-0.055 \pm 3.350i$	1.9	0.0165	12.5
Dutch Roll (Flight Sim)	$-0.350 \pm 4.760i$	1.3	0.0744	1.8
Roll Subsidence	$-3.993 \pm 0.000 i$	-	1.0000	0.2
Spiral	$-0.036 \pm 0.000i$	-	1.0000	19.4

Looking at Figure 16 to Figure 18, oscillations seem present in all modes which shouldn't really be the case. The oscillations persist even when the stability derivatives are changed, indicating that there may be some problem within the Simulink model itself. Due to project time constraints, the matter was not investigated further. Despite the oscillations, Figure 17 shows that the roll subsidence mode does return the roll angle and roll rate to zero as expected since the mode is stable. Another noteworthy observation is that the yaw angle increases significantly during roll subsidence indicating that the spiral mode may be triggered too. Figure 18 shows that the roll angle returns to zero indicating that the mode is stable.

4.2.3 Varying cruise velocity

Calculation of lateral stability derivatives is quite a tedious process and involves using empirical data from the ESDU, so it is not easy to automate and repeat for many cases. The lateral stability derivatives were thus assumed to remain constant during cruise, regardless of the cruise velocity. This assumption is justified as generally speaking the Lateral Stability derivatives depend on aircraft design characteristics (such as sweep, taper ratio and lift curve slope) as well as geometric parameters.

The Roll Subsidence mode is almost completely dependent on the rolling moment causing the perturbation [1], hence it is assumed that U_e doesn't significantly influence this mode. By considering the coupling between roll and yaw rates for an aircraft that is banking, an approximation for the spiral mode can be made using Equation 23 from [1]:

$$\lambda = -\frac{g}{U_e} \frac{L_V N_r - L_r N_V}{L_V N_p - L_p N_V} \quad (23)$$

The eigenvalue for the dutch roll mode can also be approximated using Equation 24 also from reference [1].

$$\lambda = \frac{1}{2} \left(\frac{N_r}{I_{zz}} + \frac{Y_v}{m} \right) \pm i \sqrt{\frac{U_e N_v}{I_{zz}}} \quad (24)$$

In order to test the validity of these approximations for the UAV, the spiral and dutch roll mode eigenvalues were approximated using Equation 23 and Equation 24 for the aircraft cruise velocity of $U_e = 58.3$ m/s. These were compared to the respective eigenvalues presented in Table 7 and it was found that the spiral mode eigenvalue approximation differed to the true value by only 18 %. The dutch roll mode approximation was poor however, with a percentage difference of 870 % for the real part. There is thus a high amount of uncertainty in the dutch roll varying velocity calculations

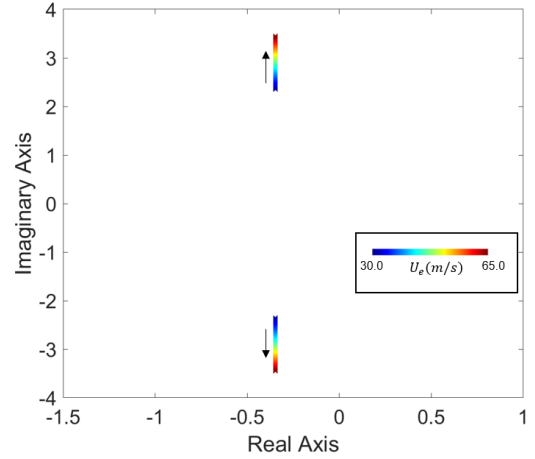
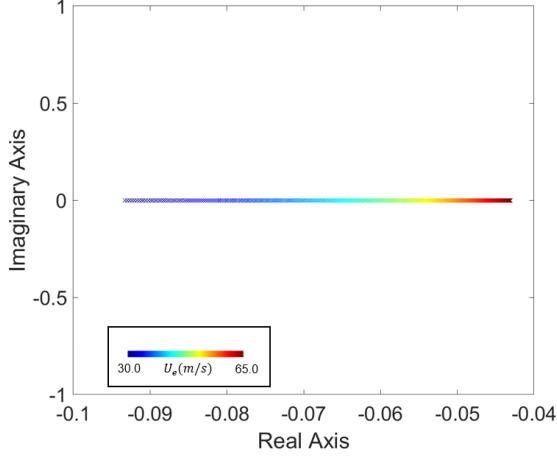


Figure 19: Root Locus Diagram varying U_e (Spiral) Figure 20: Root Locus Diagram varying U_e (Dutch Roll)

It is evident from Figure 19 and Figure 20 that neither mode can be rendered unstable by changing the velocity between the stall speed and the maximum speed. The damping ratio of the dutch roll mode can be increased if the speed is decreased. This may be worth considering in future iterations as the flight simulator team found that the damping ratio of the dutch roll mode was very small.

5 Conclusions

In conclusion, based on the linearized state-space model, all of the open loop longitudinal and lateral dynamic modes have been shown to be stable. Furthermore, key design parameters that affect the dynamic stability such as the centre of gravity location, fuel weight and cruise velocity were all varied and taken into consideration during the aircraft design process. In particular, these analyses put an important constraint on the centre of gravity location for the weight and balance team. The Dynamic stability analysis also validated key design choices such as whether or not to use anhedral or sweep as both of these parameters affect the lateral stability. A few non-standard design features such as the H-tail empennage and large protruding engine mounts were all given the green light as far as dynamic stability is concerned.

Once the flight tests in the department simulator had been conducted there were found to be several discrepancies. These were primarily with predicted damping ratio's and can largely be attributed to modelling assumptions discussed in Section 3 but also factors such as the increased attached flow over the wings due to the pusher propeller, inaccurate static margin predictions as well as flight testing the aircraft at different conditions to its design. Overall, the following recommendations are made for the next iteration of the design process:

1. The design cruise velocity could be increased. This will have the effect of increasing the damping ratio of the Phugoid mode and could possibly make the aircraft more controllable open loop. This affect may cause increased dutch roll oscillations but the increased performance in longitudinal open loop stability is perceived to outweigh this. One drawback of this is that the wing and several other aircraft components (such as the propellers) may have to be redesigned to achieve the range required.
2. The centre of gravity location is currently acceptable but could be optimized further to give larger damping ratios. As Figure 9 illustrates, the Phugoid mode is quite sensitive to changes in cg at around $x_{cg} = 0.65$ m.
3. Further investigations into the effect of the pusher propeller on the wing and H-tail need to be considered and if the propeller force is found to vary significantly during cruise it needs to be accounted for.

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Appendices

A Experimental Circular Cylinder Data

The figure below shows the drag estimation for a circular cylinder from [5].

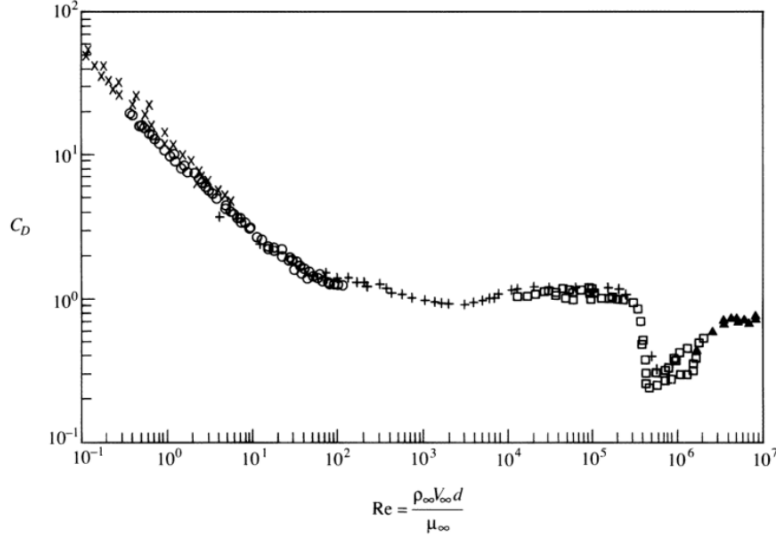


Figure 21: Experimental Data for the estimation of the drag coefficient of a circular cylinder

B Longitudinal Stability Derivative Calculations

$$X_U = \frac{1}{c_s} \left(\frac{\partial T_x}{\partial M_\infty} \right)_e - \rho U_e S_{ref} (C_D)_e - \frac{\rho (U_e)^2 S_{ref}}{2c_s} \left(\frac{\partial C_D}{\partial M_\infty} \right)_e \quad (25)$$

$$Z_U = -\frac{1}{c_s} \left(\frac{\partial T_z}{\partial M_\infty} \right)_e - \rho U_e S_{ref} (C_L)_e - \frac{\rho (U_e)^2 S_{ref}}{2c_s} \left(\frac{\partial C_L}{\partial M_\infty} \right)_e \quad (26)$$

$$M_U = -\frac{l_T}{c_s} \left(\frac{\partial T_z}{\partial M_\infty} \right)_e + \frac{\rho U_e^2 S_{ref} \bar{c}}{2c_s} \left[\left(\frac{\partial C_{M_0}}{\partial M_\infty} \right)_e - \frac{\partial K_n}{\partial M_\infty} (C_L)_e - K_n \left(\frac{\partial C_L}{\partial M} \right)_e \right] \quad (27)$$

$$X_W = \left(\frac{\partial T_x}{\partial \alpha} \right)_e + \frac{1}{2} \rho U_e S_{ref} \left[- \left(\frac{\partial C_D}{\partial \alpha} \right)_e + (C_L)_e \right] \quad (28)$$

$$Z_W = \left(\frac{\partial T_z}{\partial \alpha} \right)_e - \frac{1}{2} \rho U_e S_{ref} \left[\left(\frac{\partial C_L}{\partial \alpha} \right)_e + (C_D)_e \right] \quad (29)$$

$$M_W = l_t \left(\frac{\partial T}{\partial \alpha} \right)_e - \frac{1}{2} \rho U_e S_{ref} \bar{c} K_n \left(\frac{\partial C_L}{\partial \alpha} \right)_e \quad (30)$$

$$X_q = (x - x_{cg}) X_W + \frac{1}{2} \rho U_e S_H l'_H \left[- \left(\frac{\partial C_{D_H}}{\partial \alpha} \right)_e + (C_{L_H})_e \right] \quad (31)$$

$$Z_q = (x - x_{cg}) Z_W - \frac{1}{2} \rho U_e S_H l'_H \left[\left(\frac{\partial C_{L_H}}{\partial \alpha} \right)_e + (C_{D_H})_e \right] \quad (32)$$

$$M_q = (x - x_{cg}) \left(\frac{\partial M}{\partial W} + \frac{\partial Z}{\partial q} \right) + l_H'^2 (Z_W)_H - h_H l'_H (X_W)_H + l'_H (M_W)_H \quad (33)$$

$$X_{\dot{w}} = \frac{l_H}{U_e} \left(\frac{d\epsilon}{d\alpha} \right) (X_W)_H \quad (34)$$

$$Z_{\dot{w}} = \frac{l_H}{U_e} \left(\frac{d\epsilon}{d\alpha} \right) (Z_W)_H \quad (35)$$

$$X_{\ddot{w}} = \frac{l_H}{U_e} \left(\frac{d\epsilon}{d\alpha} \right) [-h_H (X_W)_H + l'_H (Z_W)_H + (M_W)_H] \quad (36)$$

C Lateral Stability Derivative calculations

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Table 8: Lateral Stability Derivative ESDU data sheets

Derivative	Contributor	ESDU Datasheet
N_v, Y_v	w	79006
	T2F	Adapted: 82010; 92007
	py	TM178
N_r, Y_r	w	71017 (Only for N_r)
	T2F	Adapted: 82017; 92007
	F	83026
N_p, Y_p	w	83006
	T2F	Adapted: 83006; 92007
L_v	w	80033; 73006
	T2F	Adapted: 82010; 92007
	F	73006
	P	06012
	py	TM178
L_r	w	72021
	T2F	Adapted: 82017; 92007
	ϵ	72021
L_p	w	85006

D L_v derivative calculation

The contribution ΔL_v was estimated using Figure 22 and Figure 23 from reference [11]. F is a correction factor based on the PrandtlâGlauert relation, $\Lambda_{TM} = 0$ is the tailplane sweep angle. All other symbols have been previously defined in the main body of the report.

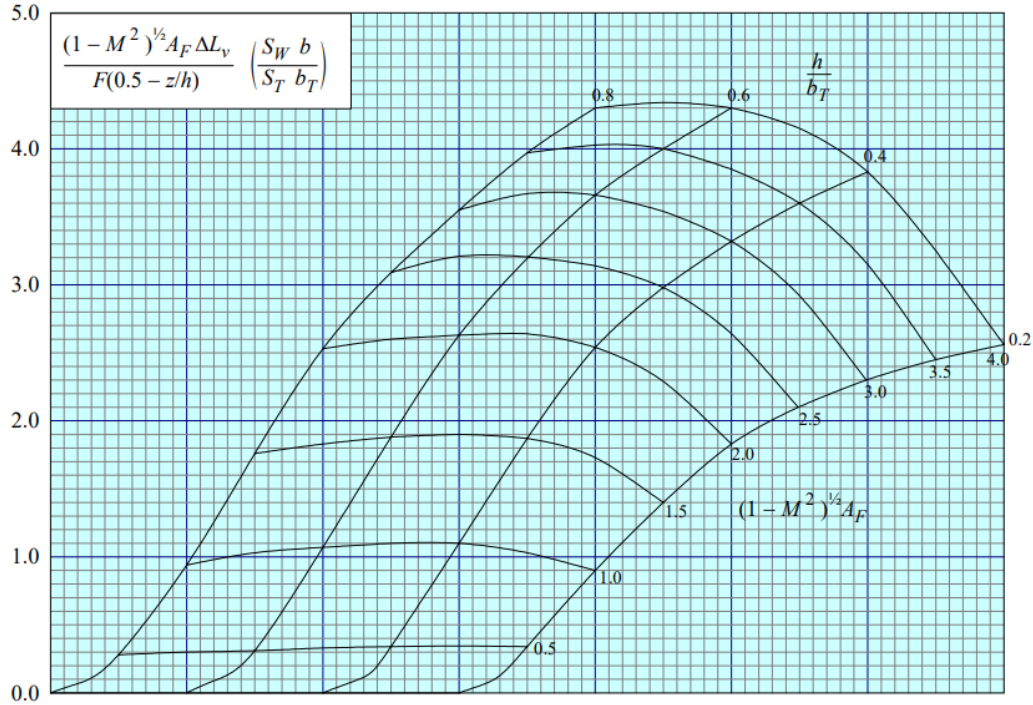


Figure 22: Calculation of ΔL_v for $(L_v)_{T2F}$ derivative

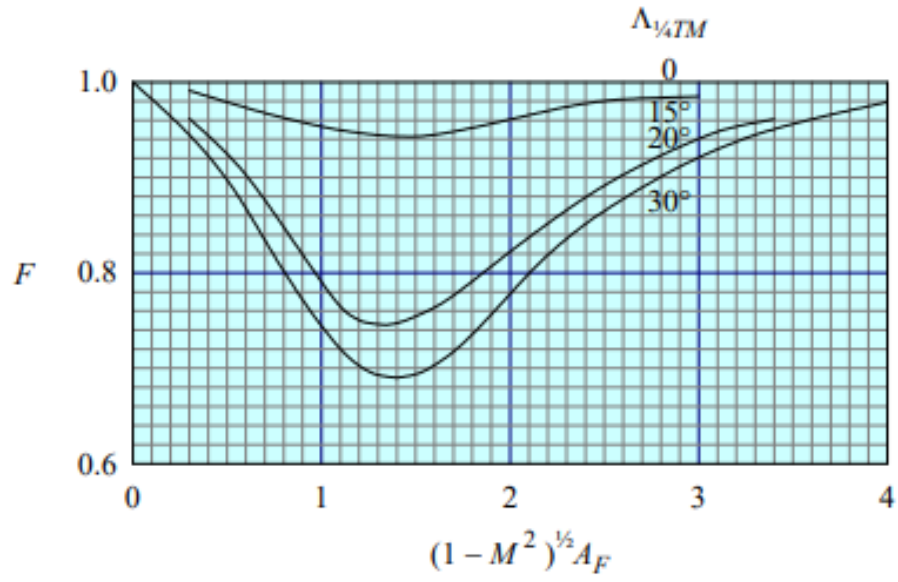


Figure 23: Calculation of factor F for ΔL_v derivative estimation

E 3D Root Locus Plot for Phugoid Mode

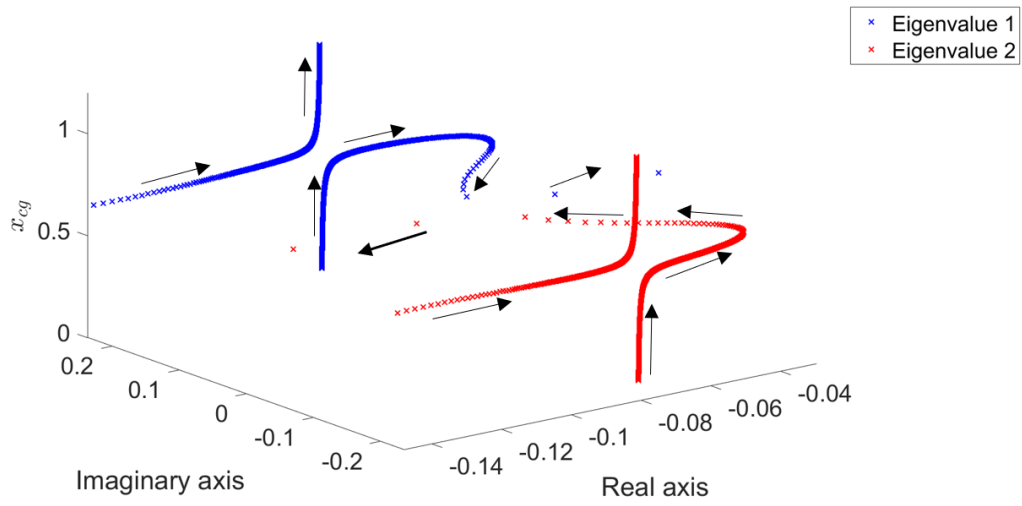


Figure 24: 3D Root Locus plot showing varying x_{cg} for the Phugoid mode